Financial Portfolio Optimization: A QAOA and VQE Formulation for Sharpe Ratio Maximization

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***Abstract*—A quantum approach for portfolio optimization involves using quantum computing to solve complex optimization problems related to portfolio selection. The approach involves constructing a mathematical model of the portfolio optimization problem using various algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) or the Variational Quantum Eigensolver (VQE). This approach has the potential to provide more accurate and efficient solutions than classical approaches, particularly for large and complex portfolio optimization problems. Moreover, this paper also dives into discussing and comparing the results obtained from both these approaches. Our work is under the assumptions required for the proposed formulation deriving meaningful considerations about the solution quality found, measured as the value of Sharpe Ratio.**

***Index Terms*—Introduction, Background, Key Concepts, Methodology, Observations, Results, Summary, Conclusion, References**

I. Introduction

Portfolio optimization refers to the selection of assets in the best possible way in order to maximize returns and minimize risks. Quantum computing helps in faster and more accurate calculations than the classical approach, therefore it can play an important role in finance and portfolio optimization. Quantum Annealing Optimization Algorithm is used to solve combinatorial optimization problems, such as the mean-variance optimization problem in portfolio optimization based on Modern Portfolio Theory. The Sharpe ratio obtained by VQE and QAOA Model is expected to yield better results as compared to the classical models as quantum computers perform better where probabilistic approach is required.

II. Background

In an optimization problem involving portfolios, each class of assets is given a weight. A weight is assigned to the assets according to a number of criteria, including average maturity, return, risk, and liquidity, etc depending upon the formulation which we are comparing.[11] Portfolio optimization techniques are fundamentally based on the principle of mean-variance maximization.[12]The field is actively investigating numerous quantum and quantum-inspired algorithms for the purpose of optimizing portfolios.For portfolio optimization, researchers are using two main computational models: Gate-based models (like IBM's devices) and Quantum Annealing (like D-Wave's systems). While Gate-based models are more flexible in terms of how they can solve problems, Quantum Annealing works well for some situations such as optimization. Nonetheless, compared to Gate-based systems, Quantum Annealing systems have been able to achieve more dependable qubits, however, these qubits encounter challenges related to low connectivity.[13] A portfolio optimization problem based on the Markowitz model for stocks traded on the Abu Dhabi Securities Exchange was transformed into a Quadratic Unconstrained Binary Optimization (QUBO) problem and then addressed making use of the Chimera architecture of the D-Wave technology and D-Wave's quantum optimizer, QBSOLV.[14] The exploration of the D-Wave system is motivated by its potential to address problems that are excessively intricate,

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time-consuming, or unresolvable through classical methods. In essence, it offers the prospect of enhancing performance relative to the existing paradigm. Consequently, it is valuable to assess how the D-Wave system stacks up against the conventional classical systems employed for solving specific problems.[15]In Recent times, We can directly extract the weights of the assets using Quantum Algorithm which was not possible earlier.

III. Key Concepts

A. Modern Portfolio Theory(MPT)

Modern Portfolio Theory constructs a diversified portfolio of investments so as to maximize returns and minimize risks. The key in MPT is diversification. Investments have two possible outcomes: high risks and high returns, or low risks and poor returns. The best outcomes, according to some, can be obtained by selecting the ideal combination of the two.

B. Markowitz Model

Markowitz is a low-risk, high-return model. The model is based on the premise that investors are reasonable and will always act in a certain way. There are only two types of assets in the Markowitz model namely: low returns and high returns.

C. Sharpe Ratio

Sharpe Ratio is the comparison of the return of an investment to its risk. By dividing excess returns by risk-free investment, Sharpe Ratio is used to determine risk-adjusted performance. The Sharpe Ratio is a crucial tool for investors since it provides a succinct evaluation of the risk-adjusted returns of an investment. It is possible to determine how effectively an asset uses risk to generate returns by comparing the excess returns of an investment (returns over the risk-free rate) to its standard deviation (a risk measurement). An improved balance between risk and return is indicated by a greater Sharpe Ratio, which helps investors create the best possible portfolio by enabling them to make well-informed decisions. This statistic is crucial for individuals attempting to handle the complexity of financial markets with prudence and precision. Maximum diversification is a portfolio approach that creates a portfolio that is as diversified as possible. Risk Parity is a technique to earn the optimal level of return at the targeted risk level.

D. Correlation

Correlation is a statistical measure that measures the degree of dependence upon the other entity. Correlation is intricately linked to the principle of diversification, which suggests that by investing in assets that exhibit low correlation, it is possible to reduce specific types of risk.

E. Quantum Approach

Quantum Computing offers computational advantages over classical computers for portfolio optimization. Quantum optimization algorithms used for portfolio optimization are Quantum Approximate Optimization Algorithm (QAOA) and Variational Quantum Eigensolver (VQE). The paper also draws comparison between the two based on the fore-mentioned factors.

IV. Methodology

A. Portfolio Optimization

Portfolio Optimization as a traditional theory was formulated by Markowitz[1]. By investing the available money, a portfolio optimization strategy seeks to improve return and reduce risk. In spite of some formulations incorporating more objectives, such as maximization of diversification and parity of risk. The formulation used here is still a constrained quadratic optimization problem; therefore, it is a general solution for testing the proposed portfolio optimization approach. A portfolio is defined as each i_{th} asset's allocation of investments X_i (based on the budget or number of units). While achieving maximum portfolio returns is often a desired objective in portfolio optimization, it is not the sole objective that investors may have. Different investors may have different priorities, such as minimizing risk or maximizing diversification. Thus, determining an optimal portfolio strategy involves weighing the trade-off between risk and return, and considering factors such as individual goals, risk tolerance, and investment horizon. There are multiple approaches that can be taken to achieve this particular objective and the paper talks about 2 of them namely: Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE). As such, investors should tailor their portfolio strategy to their unique needs and objectives.

$$\min_{x \in \{0,1\}^n} q x^T \Sigma x - \mu^T x$$
$$\text{subject to: } 1^T x = B$$

Fig. 1. Equation

Here in the equation:
The binary vectors x[i] = 1 and x[i] = 0 represent the assets to choose and not to choose, respectively. μ details the anticipated return on the assets. $\sum$ specifies the co-variances among the assets. Decision-maker B provides the budget, or the number of assets to be chosen from n, and q ¿ 0 is the control of the risk on that decision-maker

B. Classical solution

The branch and bound method is a classical optimization algorithm used to solve complex problems, particularly in multi-factor portfolio selection. This methodology uses a lot of techniques to achieve the same. We are using SLQP that is Sequential Least Squares Programming. It employs Lagrangian dual relaxation and continuous relaxation to transform a discrete optimization problem into an integer quadratic programming problem, which is then solved using a branch-and-bound approach.This approach incrementally modifies the weights given to various assets to make sure they respect

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limitations like budgetary restraints and minimum/maximum allocation caps. SLSQP assists in building portfolios that strike the ideal balance between risk and return, aligning with an investor's unique objectives and risk tolerance, by skillfully balancing these factors. The SCIPY library is a powerful implementation of the branch-and-bound algorithm, providing a robust solution to many optimization problems.

C. Quantum formulation

On near-term quantum computers, a popular technique for performing quantum chemistry calculations is the Variational Quantum Eigensolver (VQE). In this method, the ground state of a particular molecule is trained on a quantum computer using the Ritz variational principle. The VQE algorithm takes as inputs molecular Hamiltonian's and parameterized circuits that set up the molecule's quantum state. In VQE, the trial state defines the cost function, which is then sent to the quantum computer to calculate the Hamiltonian expectation value. The best estimate of the ideal parameters is then updated using the measured value to calculate the cost function's gradient.[8]

Combinatorial optimisation issues are tackled by the hybrid

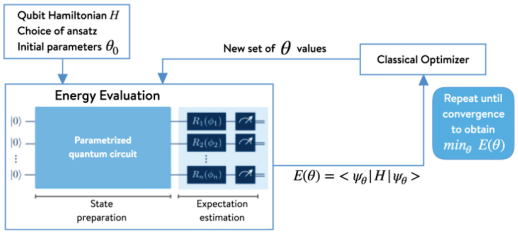

Fig. 2. VQE

variational quantum-classical technique known as the quantum approximation optimisation algorithm (QAOA). Though evidence suggests that the typical QAOA Ansatz in its fixed form is not optimal, there is currently no systematic method to find better Ansatze (An "Ansatz" is a particular shape or structure of a quantum circuit that is selected as a starting point for resolving a specific issue in the context of quantum computing and quantum algorithms) .This obstacle is overcome by creating an iterative, problem-specific form of QAOA that can also be tailored to certain hardware constraints. We evaluate the method on a class of Max-Cut graph problems and demonstrate that it converges significantly faster than the classical QAOA, despite requiring fewer CNOT gates and optimization settings. We show that the idea of shortcuts to adiabaticity is connected to this acceleration.

V. Observations

Using the Scipy library in Python, portfolio optimization can be performed based on mean-variance optimization, a technique also known as Modern Portfolio Theory (MPT). The portfolio can be optimised by using MPT in one of two ways:

- Either to minimise risks with a particular level of return
- Or to maximise returns with a certain level of risk.

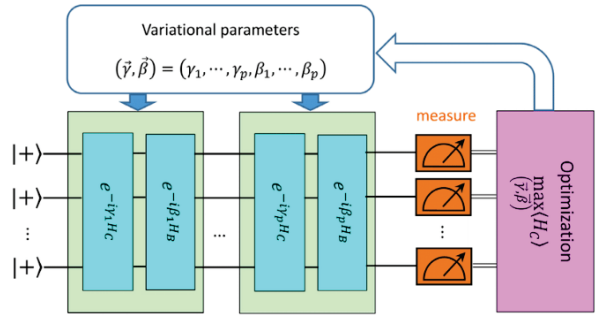

Fig. 3. QAOA Formulation

For each sample, we ran our operations on 10 stocks, and the highest Sharpe Ratio we could achieve was 1.14. Portfolio optimisation can also leverage quantum computing techniques as the Quantum Approximate Optimisation Algorithm (QAOA) and Variational Quantum Eigensolver (VQE). These algorithms take advantage of the capabilities of quantum computing to optimize portfolios depending on certain goals and restrictions, and the Qiskit library offers tools for portfolio optimization using these methods. When determining the best portfolio weights to maximise returns or reduce risk, VQE and QAOA can be utilized. In some circumstances, these methods can produce better results than traditional optimization techniques. We have performed an experiment based on Risk Factor. We can increase or decrease Risk factor manually based on our requirements. We performed our operations on 10 stocks in each sample and got maximum Sharpe Ratio upto 1.20. Hence, it is 60 base points more than the classical approach.

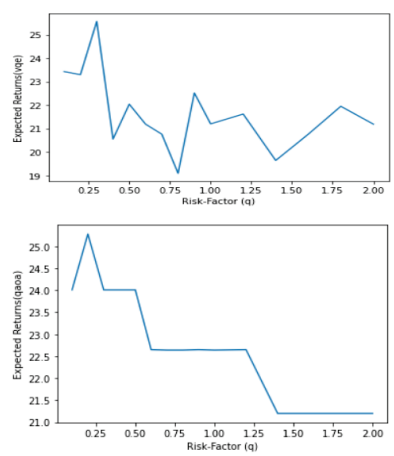

Fig. 4. Expected Returns using QAOA and VQE

*1) **Maximize Sharpe Ratio**:* In this study, quantum computing is used to solve the Portfolio Optimisation problem. In particular, a quantum approximate optimisation algorithm (QAOA), which is equivalent to the ideal portfolio, is created

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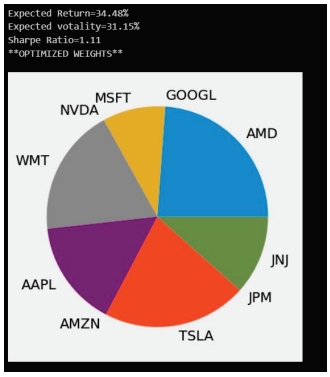

Fig. 5. Optimized Weights in Classical Approach

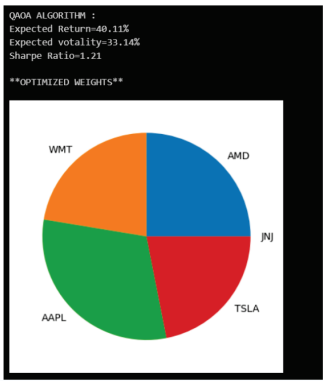

Fig. 6. Portfolio Optimization using QAOA

by translating a highly general quadratic problem simulation.Different aspects of this technique are specifically examined, including the optimizer utilised in VQE and QAOA as well as the risk coefficient, which converts confined to unconstrained issues. Additionally, investigations were carried out employing quantum simulators.

The Figures below show the Sharpe Ratio graph obtained while using these sets of Stocks ”AMD”, ”GOOGL”, ”MSFT”, ”NVDA”, ”WMT”, ”AAPL”, ”AMZN”, ”TSLA”, ”JPM”, ”JNJ” [Fig. 7 vs Fig. 8] The Figures below show the

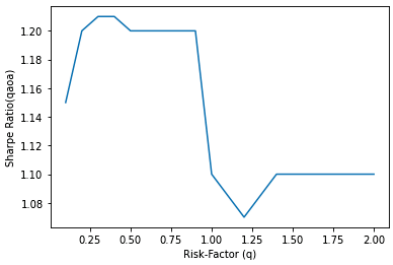

Fig. 7. Sharpe Ratio obtained using QAOA

Sharpe Ratio graph obtained while using these sets of Stocks ”META”, ”UNH”, ”XOM”, ”PG”, ”MA”, ”HD”, ”CVX”, ”ABBV”, ”MRK”, ”AVGO” [Fig. 9 vs Fig. 10]

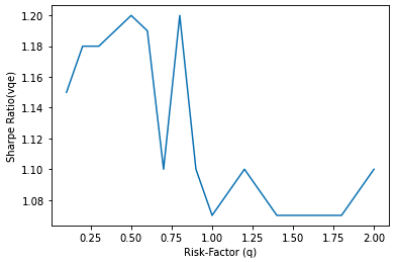

Fig. 8. Sharpe Ratio obtained using VQE

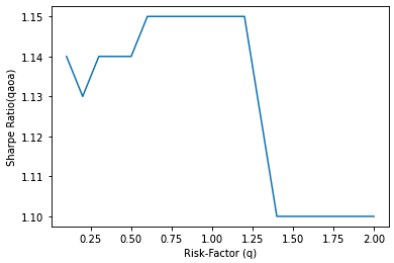

Fig. 9. Sharpe Ratio obtained using QAOA

A. *Volatility*

Volatility in portfolio optimization refers to the degree of variation or fluctuation in the returns of a portfolio’s investments over a specific period of time. It is a critical metric for investors as it provides insight into the potential risks associated with a particular investment or portfolio. High volatility implies greater price swings, indicating a higher level of risk, while low volatility suggests more stable and predictable returns.

In portfolio optimization, managing volatility is paramount. Investors seek to strike a balance between risk and return by constructing portfolios that aim to achieve the highest possible return for a given level of volatility, or conversely, the lowest possible volatility for a desired level of return. This is often done through diversification, a strategy that involves spreading investments across different asset classes or securities. By combining assets with different risk-return profiles, investors aim to offset the impact of adverse price movements in one asset with positive movements in another.

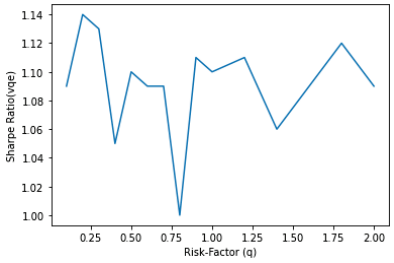

Fig. 10. Sharpe Ratio obtained using VQE

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The Figures below show the volatility graph obtained while using these sets of Stocks ”AMD”, ”GOOGL”, ”MSFT”, ”NVDA”, ”WMT”, ”AAPL”, ”AMZN”, ”TSLA”, ”JPM” [Fig. 11 vs Fig. 12]

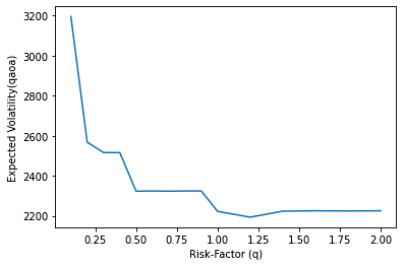


Fig. 11. volatility obtained using QAOA

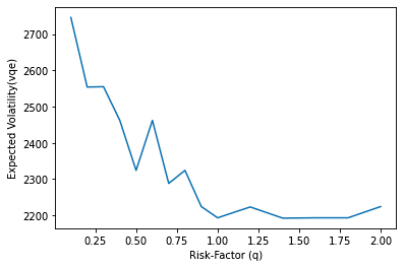


Fig. 12. volatility obtained using VQE

The Figures below show the volatility graph obtained while using these sets of Stocks ”META”, ”UNH”, ”XOM”, ”PG”, ”MA”, ”HD”, ”CVX”, ”ABBV”, ”MRK”, ”AVGO” [Fig. 13 vs Fig. 14]

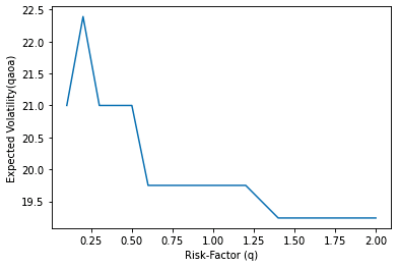


Fig. 13. volatility obtained using QAOA

B. Correlation

This concept is particularly crucial in the realm of finance and investments, where it is intricately connected to the principle of diversification. Diversification is a strategy that aims to spread risk across different assets or investments to minimize the impact of any single investment’s poor performance.

Correlation plays a pivotal role in diversification by providing insights into how various assets move in relation to one another. When assets have a low correlation, it means they tend to move independently or in different directions. This is

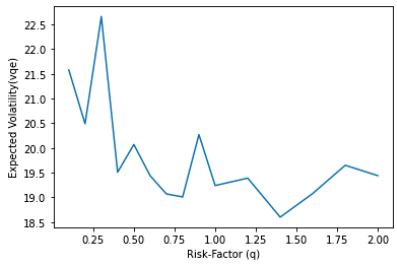
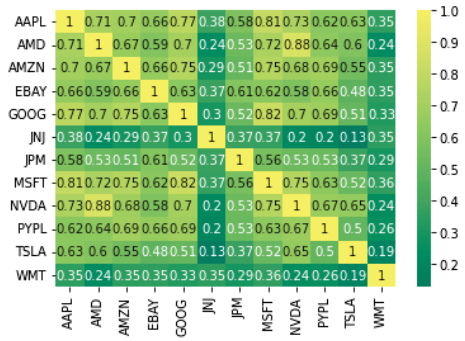
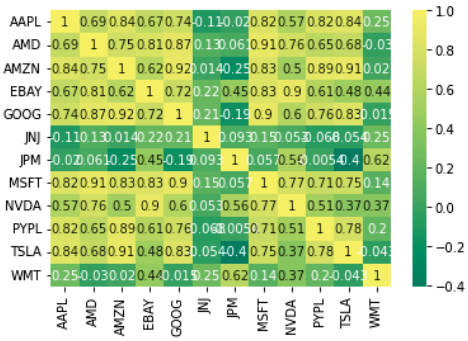


Fig. 14. volatility obtained using VQE

advantageous for investors because it implies that when one asset is experiencing a downturn, another may be performing well, potentially offsetting losses and reducing overall risk.

Conversely, assets with high correlation tend to move in tandem. In such cases, diversification might be less effective in mitigating risk since a decline in one asset could likely be mirrored by others. Therefore, understanding and analyzing correlation is a critical tool for investors seeking to construct well-balanced portfolios that can weather various market conditions. It allows them to strategically allocate resources among assets with the aim of optimizing returns while minimizing exposure to undue risk.





The study here then continues towards drawing comparisons between both these approaches that is QAOA and VQE and also providing a insightful analysis between the two.

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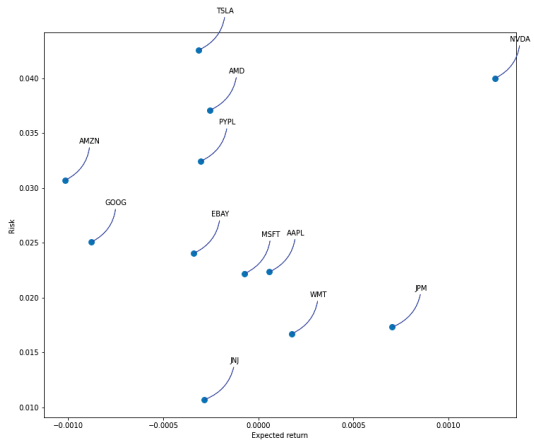

Fig. 15. Corelation Analysis Conclusion

VI. Challenges

Restricted qubit count and connection of near-term quantum devices - More qubits are required to represent the optimization problem as the number of assets rises. The qubit counts of current devices are restricted to a maximum of 20 qubits. Furthermore, qubit connectivity is restricted, which complicates the mapping of more complex problems.Errors and noise: The precision of results is decreased by errors resulting from the noisy nature of current quantum technology. Noise has an exponentially larger impact as problem sizes increase. It will require quantum error correction prior to widespread adoption.Complexity of problem encoding: At bigger sizes, portfolio optimization problems can become extremely complex when mapped into quantum hardware. It will be crucial to have effective encoding strategies that reduce the quantity of qubits needed.

VII. Results

Finally, the solutions obtained were compared to the classical solution as a benchmark. The Sharpe ratio obtained by VQE Model and QAOA Model is 1.20 and 1.21 respectively which is better than the one obtained from the classical model having a value of 1.11. Looking forward, there is potential to apply these methods to solve real-life portfolio optimization problems with larger market sizes once quantum devices with suitable characteristics become available. Additionally, future work aims to formalize the problem using the most advanced methods currently available.

	Classical Model	Quantum Models	
	Scipy	VQE	QAOA
Expected Returns	34.48	45.27	40.11
Expected Risk	31.15	37.69	33.14
Sharpe Ratio	1.11	1.20	1.21

Fig. 16. Comparison between Classical and Quantum approach

VIII. Future Works

Alternative methods for encoding problems to reduce the amount of qubits needed.Techniques for reducing noise, include reduced circuit depths and error-correcting systems.Validating encodings on quantum simulators prior to executing on quantum hardware.To further advance portfolio optimization implementations, we also intend to work with quantum hardware providers to obtain next-generation quantum computers with better qubit counts, connectivity, and noise qualities.Some Error mitigation techniques are also required.

IX. Summary

QAOA that is Quantum Approximate Optimization Algorithm can be considered to be the algorithm for solving (approximately) a subset of combinatorial problems, and VQE to be a subroutine of QAOA which might not necessarily be as in MaxCut. The Variational Quantum Eigensolver(VQE) solves the problem of finding the minimum eigenvalue of a Hamiltonian operator **H**. As a corollary, we also get a classical description of approximate ground state.It does so by classically varying over efficiently preparable ansatz states

$$|\psi(\theta)\rangle$$

and a quantum subroutine determines the expectation value

by a sampling procedure. QAOA's cost function where the repeatedly until you statistically converge enough to pick the appropriate computational basis state that will encode the solution with a high probability. Note: Due to the optimization routine above, the state should have a high overlap with its eigenstate, i.e., the lowest eigenvalue, which we emphasize once again, is the basis vector of the calculation basis.The

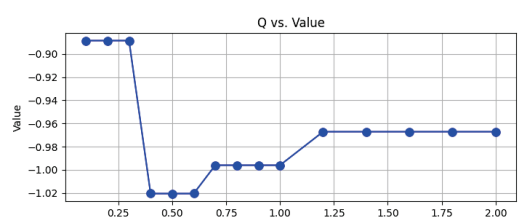

Fig. 17. Figure 1: Results Received using QAOA

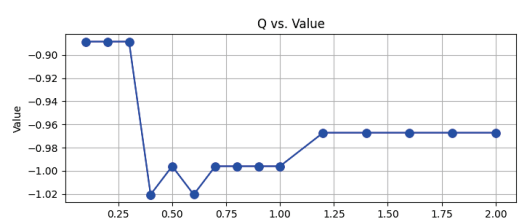

Fig. 18. Figure 1: Results Received using VQE

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$$C_i(z)$$

$$\mu = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

comparison between the two can be clearly seen taking the help of the two graphs calculated based on taking 4 stocks.We Observed in Q¡1 generally, both the algorithm behaves different for some of the Q values.The 0.09 and 0.10 Sharpe Ratio improvements for VQE and QAOA, respectively, show improvements over classical methods for portfolio optimization that are practically meaningful. Even slight increases in the Sharpe Ratio have a significant impact on quantitative finance, despite the improvement's apparent tiny absolute magnitude. Optimizing the Sharpe Ratio is closely related to the portfolio's risk-adjusted returns. Therefore, even small gains in this area can result in significantly larger returns for the same level of risk over extended periods of time.The experiments were also carried out on a modest portfolio of 10 stocks. The Sharpe Ratio uplifts are anticipated to compound even more for larger portfolio sizes, which are more typical in real-world financial applications.Both the QAOA and VQE algorithm implementations mapped the portfolio optimization problem of maximizing Sharpe Ratio onto qubit representations using standard formulations. Using QAOA and VQE hybrid quantum-classical algorithms, the problem's objective function, constraints, and decision variables were converted into Hamiltonians and ansatz states for variational optimization. Various sets of starting parameters were examined throughout the variational optimization process.All things considered, the findings show that both quantum algorithms can improve portfolio optimization results. More research will help reveal deeper understandings of financial portfolio construction as quantum processors progress toward reduced noise levels and larger scale.

CONCLUSION

The Portfolio Optimization problem is a well-known task in Financial Services and has recently drawn attention within the Quantum Computing literature thanks to the applicability of Quantum Annealers to solve the problem. In this work we tackle a specific strategy to find the optimal allocation of investments over a set of assets, namely the Sharpe Ratio maximization. We have highlighted its benefits and potential drawbacks and proposed a novel optimization algorithms to address the fore-mentioned drawbacks. We run our experiments on classical and Quantum Computing Financial Portfolio Optimization: for Sharpe Ratio Maximization [10] hardware and compare our results both in terms of the Q formulation and in terms of the optimization solver and Classical SLQP solver. **We found that Quantum algorithms are giving better results than classical solver.**

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